Experiment No-9 : Travelling Salesman Problem

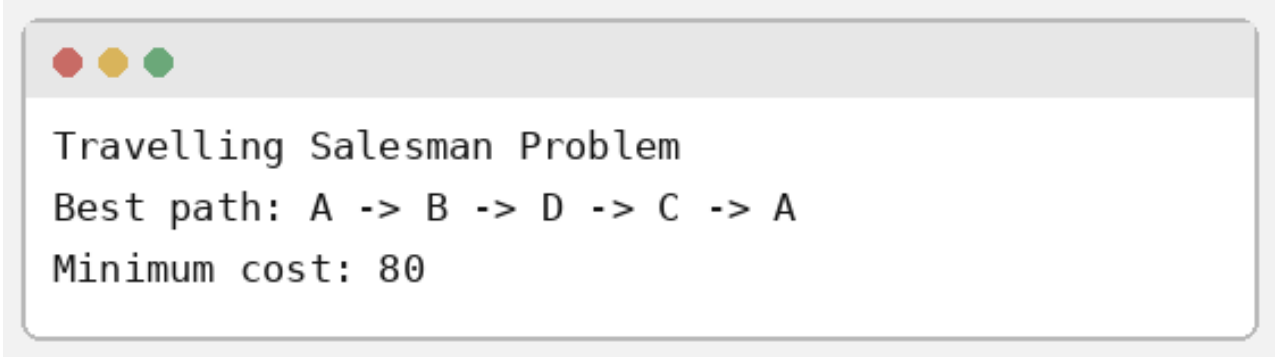
Aim

To write a Python program to implement the Travelling Salesman Problem in Artificial Intelligence.

Programming Code:

```
from itertools import permutations

def tsp(graph, start):
    cities = [city for city in graph if city != start]
    best_path = None
    best_cost = float("inf")

    for order in permutations(cities):
        path = [start] + list(order) + [start]
        cost = sum(graph[path[i]][path[i + 1]]
                   for i in range(len(path) - 1))
        if cost < best_cost:
            best_cost = cost
            best_path = path

    return best_path, best_cost

graph = {
    'A': {'B': 10, 'C': 15, 'D': 20},
    'B': {'A': 10, 'C': 35, 'D': 25},
    'C': {'A': 15, 'B': 35, 'D': 30},
    'D': {'A': 20, 'B': 25, 'C': 30}
}

path, cost = tsp(graph, 'A')
print("Travelling Salesman Problem")
print("Best path:", " -> ".join(path))
print("Minimum cost:", cost)
```

OUTPUT:



Result

Thus, the Travelling Salesman Problem was implemented successfully in Python. The program checks possible city orders and displays the minimum-cost tour.